

VIKAS SHUKLA

Sr. Software Engineer • Machine Learning Systems • Full-Stack SWE

804-386-9533 | vik.shukla44@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Bellevue, WA

Sr. Software Engineer with 7+ years of experience building scalable, high-performance distributed systems. Proven technical leader with a strong track record of delivering innovative solutions, leading cross-functional teams, and driving measurable business impact.

EXPERIENCE

Sr. Software Engineer | Capital One | Bellevue, WA

May 2023 – Present

Cards Platform Engineering

- Architected and deployed cloud-native REST microservices supporting real-time fraud decisioning across 40M+ transactions with low-latency, high-availability production deployments.
- Built production observability and telemetry framework including metrics, structured logging, dashboards, alerting, automated health checks, and data validation, reducing production incidents by 80%.
- Designed scalable backend services and data pipelines integrating relational and NoSQL data sources into a unified production platform serving multiple engineering teams.
- Designed and implemented cloud-native platform services, shared libraries, and deployment pipelines supporting multiple engineering teams, improving platform scalability, maintainability, and developer productivity.
- 60% coding & implementation, 10% system design (HLD; LLD), 25% teams operational excellence(on-call) and 5% hiring/mentoring.

Software Engineer | Amazon | Bellevue, WA

Jun 2022 – Apr 2023

- Designed and deployed secure machine learning development infrastructure supporting model training, experimentation, and deployment workflows for the Alexa Data Science organization.
- Built AI-powered notification quality assessment framework integrating user settings, reachability signals, and A/B test results across push, SMS, and email channels; improved Alexa user engagement and retention by 20%.
- Launched uplift-based targeting system with guardrails against notification fatigue, reduced total sends by 22% with no engagement loss, cut SMS/email failures by 30%, and lifted incremental conversion by 8%.
- Built scalable REST APIs and backend services following object-oriented design principles, supporting production experimentation platforms and distributed data pipelines.

Software Engineer | SGS Teknics | Gurugram, India

Nov 2014 – May 2018

- Automated on-premises server fleet scaling for SGS tax engine during peak holiday traffic, reducing deployment time from 3-4 hours per server to just 25 seconds.
- Led the infrastructure design and deployment of SGS tax processing platform, ensuring high availability, scalability, and reliable performance during peak sales periods.
- Mentored interns, organized innovation hackathons, and delivered technical training programs to strengthen team capability and engineering excellence.

KEY PROJECT

IntelliQuery Engine [🔗](#) | Python, FastAPI, FAISS, React Native, SQL

- Implemented agentic routing and context-aware retrieval workflows for finance/legal queries, improving accuracy by 45% and reducing manual validation effort by 50%.
- Deployed scalable FastAPI backend integrated with React Native application ([caseFit](#)) delivering production-grade low-latency AI responses.

EDUCATION

M.S Information Studies & AI | Trine University, Reston, VA

Feb 2025

M.S. Data Science (GPA: 3.7 / 4.0) | University at Buffalo

Feb 2022

B.Tech. Electrical & Electronics Engineering (GPA: 8.6 / 10) | SRM University, India

May 2014

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, C++, SQL, TypeScript, Bash

Engineering: Object-Oriented Programming, Scalable Architecture, Distributed Systems, Microservices

HLD, LLD, REST APIs, Data Structure & Algorithms, Multithreading, Linux, Production Troubleshooting, Git

Cloud & DevOps: AWS Services (Lambda, S3, ECS Fargate, DynamoDB, RDS, SQS, SNS, Kinesis, CloudFormation, Glue EventBridge, AppConfig), Azure Services (Cosmos DB, Kafka, Kubernetes)

Frameworks: .NET, ASP.NET, FastAPI, React JS, React Native, Flask